another.This repulsive force is called zeta potential. Because of its size and pentameric in nature, IgM can overcome the zeta potential and cross link red blood cells, leading to agglutination. The smaller size and bivalency of IgG makes it less able to overcome the zeta potential. For this reason, IgM is more effective than IgG in agglutinating red blood cells.

Bacterial Agglutination (Tube Agglutination)

A bacterial infection often elicits the production of serum antibodies specific for surface antigens on the bacterial cells. The presence of such antibodies can be detected by bacterial agglutination reactions.

Widal test is used for the diagnosis of typhoid fever. In typhoid patients, the serum contains antibodies to *Salmonella typhi*. In Widal test, two antigens are used. They are antigen H, the flagellar antigen and O antigen, the somatic antigen. When antiserum of the patient is added to the antigens, the antigens are clumped and identified.

Antiglobulin Test (Coombs Test):

The antiglobulin test was devised by Coombs, Mourant and Race for the detection of anti-Rh antibodies that do not agglutinate Rh positive erythrocytes in saline. When sera containing incomplete anti-Rh antibodies are mixed with Rh positive red cells, the antibody globulin coats the surface of the erythrocytes, though they are not agglutinated. When such erythrocytes coated with the antibody globulin are treated with a rabbit antiserum against human gammaglobulin (antiglobulin or Coombs serum), the cells are agglutinated. This is the principle of the antiglobulin test.

10.7. BLOOD GROUPS

A blood type is a classification of blood, based on the presence and absence of antibodies in blood and inherited antigenic substances on the surface of red blood cells. These antigens may be proteins, carbohydrates, glycoproteins, or glycolipids, depending on the blood group system.

10.7.1. ABO System

Karl Landsteiner discovered ABO blood group system. He was awarded Noble Prize in 1930 for his discovery. The membranes of human red cells contain a variety of blood group antigens,which are also called **agglutinogens.** The most important and best known of these are the A and B antigens. Antibodies against red cell agglutinogens are called **agglutinins.** Antibody-A and Antibody-B are examples of agglutinins. Landsteiner law states that if an agglutinogen is present on the RBC of an individual, the corresponding agglutinin must be absent in the plasma of that individual and vice-versa.

There are four main blood groups defined by the ABO system:

Blood group A people have A antigens on the red blood cells with anti-B antibodies in the plasma. Blood group B people have B antigens with anti-A antibodies in the plasma

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